Mathematical Methods in Uncertainty Quantification (MMUQ) – L00

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MMUQ L00 – Introduction

- Goal(s) of this Course
- Structure
- Requirements and Expectations
- Assignments
- Groups

Goal(s)

- Uncertainty quantification in general
  - of the model output(s) mainly
- Model parameter optimization
- Parameter sensitivity
- Input uncertainty
- Output uncertainty
- Topics/direction for your Master's Thesis

Structure

- Seminar
  - Main part
  - Some theory presented
  - Assignments presented
  - Interdisciplinary
  - Presentations/Discussions
- MOOC
  - Massive Open Online Courses
  - Independent completion of assignments
  - Groundwater
- Tuesdays in person
- Fridays online
- Seminar and MOOC interlinked
- Final report
  - Seminar results
  - MOOC results
  - All code
- Final report presentation
  - Summarize seminar highlights
- Both courses are 9 ECTS combined

Requirements and Expectations

- Background
  - Hydrology
  - Mathematics
  - Computer Programming
- Interdisciplinary interactions
  - Explain to each other
- Working in a group
  - Nicht im Stich lassen!
- Windows!
  - Portable IDE and Python supplied
  - Input data provided
- Eagerness to explore
- Embrace uncertainty
  - Organization of this course!
  - Don’t go into too much detail!
- Complete assignments on time, please
- You do most of the work
  - This is a seminar
  - I only guide and control
- Assignment presentations on Fridays (normally)

All data/software on Webdisk

Log in to: https://webdisk.ads.mwn.de/

Navigate to:

- TUBV → hfm → public → MMUQ_WS2526
- For downloading:
  - Right click on a directory/file
  - Left click Download

Assignment – 1

- Model parameter optimization (Differential Evolution)
- High flow event(s)
- Get a feeling of how model parameter optimization works
- Various objective functions
- Presentation
  - Convergence
  - Each parameter vs. objective function (OF)
- Reference: Long-term lumped projections of groundwater balances in the face of limited data, Ejaz 2024, Ph.D. Thesis

Assignment – 2

- Local sensitivity analysis (SA) of model parameters
- Effects of each parameter on the OF
  - Relative perturbation
  - Quantity of influence on model accuracy
- Use of a calibrated model
- Presentation
  - Each parameter vs. OF
  - Uncertainty bounds of model output
  - Sensitive parameter ranking
- Reference: Uncertainty quantification, Smith 2014, Book

Assignment – 3

- Global SA of model parameters
- Combined effect of parameters on the OF
- Sampling in parameter space
- Various methods!
  - Sobol
- Presentation
  - Uncertainty bounds of model output
  - Sensitive parameter ranking
- Reference: Uncertainty quantification, Smith 2014, Book

Assignment – 4

- Model output uncertainty due to uncertain inputs
- Perturb inputs conditionally
- Analyze effects on model output
- Presentation
  - Make comparisons
    - Calibrated model output
    - Recalibrate model
  - Uncertainty bounds of model output
- Reference: Impact of biased and randomly corrupted inputs on the efficiency and the parameters of watershed models, Oudin et al. 2006, Paper

Assignment – 5

- Considering model output uncertainty
- Limits of acceptability
- Presentation
  - What happens to model performance?
  - Uncertainty bounds of model output
- Reference: Towards a limits of acceptability approach to the calibration of hydrological models - Extending observation error, Liu et al. 2009, Paper

Groups

- Mix of disciplines
- Two or three members?
- Examples →

	Math.	Envi.	Mix
A	Jonas Plank	Roman Ramsperger	Yufei Wang
B	Zhizhi Jing	Christine Leers	Yihan Shen
C	Kristi Nita	Umaima Jalil	Ding Wang
D	Adrien Thébault		Xiaoyu Zhang

That was it for today, thanks!